DineDesk

Online Restaurant Ordering & Table Reservation Platform — Full Project Requirement

Project Overview and Discussion

Project Theme

DineDesk is a digital platform that connects food lovers (customers) with restaurants and home-kitchens (vendors). Customers can browse restaurants, explore menus, order food for delivery, reserve tables, and pay online. Restaurant owners can list their business and manage menus after a one-time verification payment, while an Admin and a Super Admin oversee the entire system.

Why should we develop the project?

Traditional food ordering is fragmented across phone calls and third-party apps with high commissions. An in-house online marketplace gives restaurants direct access to customers, provides a secure and streamlined ordering + reservation experience, and demonstrates advanced MERN stack concepts including multi-level role-based access, payment integration, real-time order tracking, and AI-assisted recommendations.

How does the system work?

- Customers browse restaurants/menus, add items to cart, checkout via Stripe or Cash on Delivery, reserve tables, leave reviews on food they have ordered, and track order/reservation history.
- Vendors (Restaurant Owners) pay a one-time publishing fee, then create and manage their restaurant profile and menu items. They can publish/unpublish, edit, delete menu items, and manage incoming orders and reservations.
- Admin manages users (role changes), approves/rejects/unpublishes restaurant listings, and monitors all transactions.
- Super Admin (top-level authority) manages Admins themselves, controls platform-wide settings (commission rate, featured restaurants, banned users), and views system-wide analytics across all admins.
- Authentication uses JWT with email/password and Google login, including Forgot Password recovery and a "Remember Me" persistent-session option.
- Dashboards are role-specific (Customer / Vendor / Admin / Super Admin) with CRUD operations, payment flows, real-time order status, and analytics charts.

Ensure the Following things to get 100% mark (No mark will count otherwise)

- Include at least 20 meaningful commits on the client side & 12 meaningful commits on the server side with descriptive commit messages.
- Include a README file with the project name, purpose, live URL, key features, and all npm packages used.
- Secure frontend configuration keys using environment variables.
- Secure your MongoDB credentials, JWT secret, Stripe keys, imgBB key, and Gemini API key using environment variables.
- Create a design that encourages recruiters — pleasant colour contrast, proper alignment, spacing, and consistent typography. Avoid a generic/templated ("gobindo") look.
- If the project is found similar to any module / conceptual / assignment project, it will receive 0 marks and may forfeit the chance of any upcoming reward.

Deployment Guideline

- If the deployment is not okay, the project will get 0 marks and may miss the chance of upcoming rewards.
- Ensure the server works perfectly in production without CORS / 404 / 504 errors.
- Ensure the live link works perfectly and does not show errors on first landing.
- Ensure that no route throws an error on page reload.
- A logged-in user must NOT be redirected to the Login page on reloading any private route.

Layout & Page Structure

1. Navbar Requirements

- Logo / site name (links to Home).
- Navigation links: Home, Browse Restaurants, Reserve a Table, Dashboard (dropdown based on role), Login/Logout.
- Global search bar (search restaurants or dishes by name, cuisine, or category).
- Cart icon with live item count, opens a mini-cart drawer.
- Dark Mode toggle (persistent across reloads).
- Responsive mobile menu (hamburger icon).
- Active route highlighting.

2. Banner Section

Beautiful hero banner featuring appetising food photography and restaurant imagery (Slider or Carousel preferred), with a tagline and a prominent "Order Now" call-to-action.

3. Footer Requirements

- Copyright information.
- Quick links (About, Contact, Privacy Policy, Terms of Service).
- Social media icons (dummy links).
- Newsletter signup placeholder (frontend only).
- Cookie / Privacy notice link.

4. Cookie Consent & Privacy

- A dismissible cookie-consent banner on first visit, storing the user's choice in localStorage.
- A dedicated Privacy Policy page describing what data is collected and how it is used.

Authentication System

Registration Requirements

- Users can register with email + password or Google Login (OAuth).
- Required fields: Full Name, Email, Password, Confirm Password.
- After registration, the option to choose a role between Customer (User) and Restaurant Owner (Vendor).

- Email must be unique.
- A welcome email is automatically sent to the user's inbox right after successful signup (via Nodemailer / EmailJS with a dummy or sandbox SMTP service).
- Successful registration returns a JWT token and redirects to Home.

Login Requirements

- Login via email/password or Google.
- A "Remember Me" checkbox — when checked, the session/JWT persists across browser restarts; when unchecked, the session clears when the browser closes.
- "Forgot Password?" link that sends a secure, time-limited password-reset link to the user's email and allows setting a new password.
- For email/password: validate credentials, generate JWT (expires in 7 days).
- Google Login: integrate Firebase Auth / BetterAuth to handle the OAuth flow, then issue a JWT.
- Role-based redirection after login (customers to Home, vendors/admins to their respective dashboards).
- Logout clears token and client-side state.

Main Pages

Home Page

- Hero Section — Large banner with tagline "Delicious Food, Delivered Fast" and CTA button "Browse Restaurants".
- Dynamic Section — "Featured Restaurants" (fetch latest 6 restaurants from DB, random or admin-selected). Auto-refreshes on page reload.
- Animation with Framer Motion — Hero text fade-in, restaurant cards staggered reveal on scroll, hover scaling effects.
- Extra Section 1 — "Top Rated Restaurants" (display 3 restaurants with the highest average rating, using cover photo and name).
- Extra Section 2 — "Food Categories" (grid of categories: Pizza, Burger, Sushi, Desi, Chinese, Desserts, etc., each linking to the browse page with a filter applied).
- Extra Section 3 — "What Our Customers Say" (rotating carousel of top customer reviews).
- AI Recommendation Widget — a floating chat assistant (see AI Integration below) suggesting dishes based on the user's mood/craving.

Browse Restaurants Page

Allow users (customers, vendors, admins, guests) to explore, search, filter, and sort all available restaurants.

- Public Access — No login required to view the page. However, ordering, reserving, and reviewing require authentication.
- Display restaurants in a grid layout. Each card shows: cover photo, name, cuisine type, average rating, price range, and a "Closed" badge if currently outside operating hours.
- Responsive grid: 2 columns on mobile, 3 on tablet, 4 on desktop.
- Loading & Error States — Skeleton cards while fetching; friendly message when no restaurants match filters.
- Click on a restaurant card — Navigates to the Restaurant Details page.

Restaurant Details Page

Show complete information about a single restaurant and allow authenticated actions (order, reserve, review).

- Public Read-Only Preview — Anyone can view the menu and details. Ordering, reserving a table, and reviewing require login.
- Restaurant information display: cover photo & gallery (from imgBB), name, cuisine/category, description, opening hours, address (Google Maps embed), average rating & review count, status (Open / Closed).
- Full menu list grouped by category (Starters, Mains, Desserts, Drinks) with photo, name, description, and price for each dish.
- "Add to Cart" button on each dish; "Reserve a Table" button opens a modal to pick date, time, and party size.
- Customer Reviews section — star rating + comment, only visible/postable to customers who have completed an order from this restaurant.
- Loading & Error States — Skeleton loader while fetching data.

Cart & Checkout

- Persistent cart (per logged-in user) showing item image, name, quantity stepper, price, and subtotal.
- Apply a coupon/promo code for a discount.
- Choose delivery address (with Google Maps location picker) or dine-in/pickup option.
- Choose payment method: Stripe (card) or Cash on Delivery.
- Order confirmation page with an order summary and an estimated delivery time.

Dashboard Layout

Route: `/dashboard`

Shows the logged-in user's profile summary and quick stats, with a Side Navbar containing dashboard route links specific to the user's role. The necessary routes for each role are listed below.

Dashboard for Customer (Role: user)

Route: `/dashboard/user/order-history`

Order History — Table with restaurant name, items, total amount, order date, and status ("pending" (default), "preparing", "out-for-delivery", "delivered", "cancelled") updated live by the vendor. A live status stepper/progress bar visualises the current stage of the order.

Route: `/dashboard/user/reservations`

Table Reservations — List of reservation requests with restaurant, date/time, party size, and status ("pending", "confirmed", "rejected") set by the vendor.

Route: `/dashboard/user/update-profile`

Profile Management — Edit profile. An "Update Profile" button on /dashboard navigates the user to this page to update Full Name, Profile Picture, phone number, and saved delivery addresses.

Route: `/dashboard/user/reviews`

Review Management — List of the logged-in user's reviews on restaurants, in table or card-grid format, with "Edit" and "Delete" buttons (via modal or dedicated edit page).

Route: `/dashboard/user/favorites`

Favorites / Wishlist — Restaurants and dishes the user has bookmarked, with a quick "Add to Cart"/"Reserve" shortcut and a "Remove" button.

Dashboard for Restaurant Owner (Role: vendor)

Route: `/dashboard/vendor/orders`

Incoming Orders — A table of orders placed by customers, containing columns: customer name, items, total amount, order time. Each row has action buttons to move the order through its status (Accept → Preparing → Out for Delivery → Delivered) or Reject.

Route: `/dashboard/vendor/reservations`

Reservation Requests — Table of incoming table-reservation requests with customer name, date/time, and party size. Each row has Accept/Reject buttons.

Route: `/dashboard/vendor/manage-menu`

Manage Menu — Table/list of the restaurant's own menu items with actions (add, edit, delete). The vendor can update dish name, description, price, category, availability toggle, and image upload (imgBB).

Route: `/dashboard/vendor/manage-profile`

Restaurant Profile — Edit restaurant name, cover photo, cuisine type, opening hours, and address.

Dashboard for Admin (Role: admin)

Route: `/dashboard/admin/manage-users`

Manage Users — User list in a table. Each row contains name, email, role, a "Change Role" button, and a "Delete User" button, both fully functional.

Route: `/dashboard/admin/manage-restaurants`

Manage Restaurants — Table of all restaurant listings with owner name, status, and actions to approve / publish / unpublish / delete a restaurant.

Route: `/dashboard/admin/all-transactions`

View All Transactions — Table: each row contains transaction ID, customer/vendor email, amount, and date.

Route: `/dashboard/admin/analytics`

Analytics Overview — Cards: total users, total vendors, total orders, total revenue, plus a monthly revenue chart.

Dashboard for Super Admin (Role: superadmin) — New Role

The Super Admin sits above the Admin and has full platform authority, including the power to manage Admin accounts themselves — a capability regular Admins do not have.

Route: `/dashboard/superadmin/manage-admins`

Manage Admins — Table of all Admin accounts with actions to promote a user to Admin, demote an Admin, or permanently remove an Admin account.

Route: `/dashboard/superadmin/platform-settings`

Platform Settings — Configure site-wide values such as commission percentage per order, featured restaurants on the Home page, and global maintenance-mode toggle.

Route: `/dashboard/superadmin/system-analytics`

System-Wide Analytics — Aggregated charts across every Admin's activity: platform growth, total GMV, top-performing restaurants, and user-growth trends.

Other Requirements

- Image Upload — Use the imgBB API for storing restaurant covers, menu item photos, and user profile pictures.
- Payment System (Stripe) — For paying online food orders; Cash on Delivery offered as an alternative.
- Transaction History — For customers, vendors, and admins.
- AI Integration (Gemini 1.5 Flash) — A floating AI chat assistant that recommends dishes/restaurants based on the user's craving or dietary preference, answers FAQs (delivery time, refund policy), and can summarise a restaurant's reviews into a short verdict.
- Customer Review & Rating System — Star rating (1–5) plus a written comment on a restaurant, restricted to customers who have a completed order from that restaurant.
- Cookie Consent & Privacy Policy — GDPR-style cookie banner and a dedicated privacy policy page.

Additional

Loading Page

Global loading spinner and skeleton loaders across data-fetching pages.

Error Page

Custom 404 page, error boundary fallback, and toast notifications for API errors.

Dashboard UI Requirements

Responsive design, consistent colour theme, full-width dashboard layout, and interactive charts (e.g., Recharts) for analytics pages.

Challenge Requirement Guideline

1. Review System on Restaurants

- Only customers who have a completed (delivered) order from a restaurant can review it.
- The API endpoint checks the order record before allowing a review to be posted.

2. Search & Filtering

On the Browse Restaurants page, implement search and filtering by name/cuisine, price range, rating, and open/closed status.

3. Pagination

Show 6–12 restaurants per page with navigation controls.

4. Order & Payment Flow

- A customer places an order from the restaurant details page or cart; the order lands in the vendor's /dashboard/vendor/orders queue as "pending".
- The vendor accepts the order and updates its status as it progresses (Preparing → Out for Delivery → Delivered).

- Payment is captured at checkout via Stripe, or marked "Cash on Delivery" and confirmed by the vendor upon delivery.
- The customer sees the live status update on /dashboard/user/order-history without needing to refresh (via polling or a socket connection).

5. Table Reservation Flow

- A customer requests a reservation with date, time, and party size from the restaurant details page.
- The vendor accepts or rejects the request from /dashboard/vendor/reservations.
- The customer sees the confirmed/rejected status on /dashboard/user/reservations, with an email notification sent on confirmation.

6. JWT / Token Verification

Use JWT (or a token provided by BetterAuth/Firebase) to secure every API call and enforce role-specific access at the middleware level (customer / vendor / admin / superadmin).

What to Submit

- Super Admin Email & Password
- Admin Email & Password
- Links: Live link, GitHub (server & client repositories).

Optional Requirement Guideline

- Loyalty Points / Rewards — Customers earn points per order, redeemable as a discount on future orders.
- "Popular" / "Chef's Special" Badge — Auto-badge on best-selling dishes.
- Email Notifications — Dummy emails for order confirmation, reservation confirmation, and account signup.
- Dark Mode Toggle — Persistent dark mode switch (already in Navbar above, listed here as an enhancement).
- Live Order Tracking Map — Simulated delivery-rider location on a map for the order-in-transit state.